

Supplementary material for *Dynamic consequences of declining vaccination coverage in a heterogeneous world*

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1 Theoretical honeymoon time in the SIR model

We consider here how reductions in vaccination can lead to rebounds in disease prevalence in a SIR (Susceptible, Infected, Recovered) model. In particular, we will focus on the post-vaccination “honeymoon”, or how long it takes for the disease to rebound after vaccination starts declining. Note, we use the term “immunization” throughout to account for both vaccination coverage and imperfect vaccine efficacy (i.e., immunization represents those successfully immunized, not merely vaccinated).

To begin, we will consider the simplest case where (a) a population has been under a “pre-decline” vaccination regime for a sufficiently long period of time to be at equilibrium, and (b) that the pathogen is locally eliminated. Let v_0 be the immunization rate during the period of herd immunity, and let v be the immunization rate after the decline begins.

In the SIR model, we know there are two possible equilibria:

1. disease-free equilibrium, where $S^* = v_0$
2. endemic equilibrium, where $S^* = \frac{1}{\beta}(\mu + \gamma + \frac{\mu(\mu+\gamma)}{\sigma})$

Here, we only consider the first case where a population is above the herd immunity threshold, $1/R_0$, because otherwise there is no honeymoon period. We are interested in the duration of the honeymoon period, which we hypothesize will depend on a number of factors including the R_0 of the pathogen, the equilibrium number of susceptibles S^* , the birth rate μ , the immunization rate v under decline, and the timing of importations. Because we cannot control the timing of importations, we define the honeymoon period as the time it takes for $R_e = R_0 \cdot S(t) > 1$. In other words, the time it takes for R_e to cross 1 is a lower bound on the honeymoon period, where any importation after this point will start an outbreak.

We can calculate analytically the time it takes for R_e to cross 1, under assumptions of local extinction (i.e., $I = 0$) and a constant post-decline immunization rate, v . In this case, we have a differential equation for the number of susceptibles, S , which we can solve using the following steps:

$$\frac{dS}{dt} = \mu(1 - v) - \mu S \quad (1)$$

$$e^{\mu t} \left(\frac{dS}{dt} + \mu S \right) = e^{\mu t} \mu(1 - v) \quad (2)$$

$$\int e^{\mu t} \left(\frac{dS}{dt} + \mu S \right) dt = \int e^{\mu t} \mu(1 - v) dt \quad (3)$$

$$\int e^{\mu t} \frac{dS}{dt} + e^{\mu t} \mu S dt = \int e^{\mu t} \mu(1 - v) dt \quad (4)$$

$$e^{\mu t} S = (1 - v)e^{\mu t} + C \quad (5)$$

$$S(t) = (1 - v) + Ce^{-\mu t} \quad (6)$$

and solving $S_0 = (1 - v) + Ce^{-\mu \cdot 0}$ gives us $C = S_0 - (1 - v)$. Thus, we have:

$$S(t) = (1 - v) + (S_0 - (1 - v))e^{-\mu t}$$

We can then find the time t at which $R_e = R_0 S(t) = 1$, which yields

$$t = \frac{-1}{\mu} \log \left(\frac{1/R_0 - (1 - v)}{S_0 - (1 - v)} \right)$$

We can also rewrite this equation to show us the relationship between pre- and post-decline immunization levels that yield equivalent honeymoon periods. When vaccination rates are above the herd immunity threshold, $S_0 = 1 - v_0$ is the pre-drop vaccination coverage, so

$$t = \frac{-1}{\mu} \log \left(\frac{1/R_0 - (1 - v)}{(1 - v_0) - (1 - v)} \right)$$

which can be simplified to

$$v_0 = v(1 - e^{t\mu}) + e^{t\mu}(1 - 1/R_0).$$

2 Unity beta calculation

Age structured dynamics are known to be important for a range of infectious disease, and we will consider how this affects theoretical honeymoon times. In particular, we will use a theoretical quantity called “unity beta” to explore these effects [ADD REFS].

First we outline the unity beta calculation. Consider an age-structured transmission process that is governed by heterogeneous contacts between age groups, where the number of new infections in the age structured model is

$$\text{age-structured infections} = \frac{1}{N} \sum_{i=1}^n \sum_{j=1}^n \theta_{i,j} S_i I_j$$

where $\theta_{i,j}$ is the transmission rate between age groups i and j . We can then calculate a time-varying quantity, $k(t)$, called “unity beta” that ensures the number of new infections in the age-structured model is equal to the number of new infections in a non-age-structured model, where

$$\text{age-structured infections} = \frac{1}{N} \sum_{i=1}^n \sum_{j=1}^n \theta_{i,j} S_i I_j \quad (7)$$

$$= \frac{\hat{\beta}(t)}{N} \sum_{i=1}^n S_i \sum_{j=1}^n I_j \quad (8)$$

$$= \text{homogenous infections} \quad (9)$$

given that

$$\hat{\beta}(t) = \frac{\sum_{i=1}^n \sum_{j=1}^n \theta_{i,j} S_i I_j}{\sum_{i=1}^n S_i \sum_{j=1}^n I_j}$$

This ratio can be interpreted as the ratio of the number of new infections in the age-structured model to the number of new infections in the homogenous model. If $k(t) > 1$ then age-structured dynamics are fast relative to the homogenous dynamics, and vice versa if $k < 1$.

In this case, however, the homogeneous model transmission rate is determined by $k(t)$, where the transmission rate becomes the value that is necessary to generate the same dynamics as the age-structured model. However, such dynamics can yield different reproduction numbers, so we may also be interested in the relationship between age-structured and homogenous dynamics when fixing the reproduction number. This is the case we use for the main text analysis.

In particular, we contrast a number of WAIFW matrices, which on their own have different basic reproduction numbers. Thus, to enable comparison, we calculate a scalar c such that the reproduction number yields a given $R_0 = \beta/\gamma$. In the main text, we use $R_0 = 17$. Thus, we have

$$\text{age-structured infections} = \frac{c}{N} \sum_{i=1}^n \sum_{j=1}^n \theta_{i,j} S_i I_j \quad (10)$$

$$\text{homogenous infections} = \frac{\beta}{N} \sum_{i=1}^n S_i \sum_{j=1}^n I_j \quad (11)$$

Following the same approach, we can equate these two quantities with a ratio $k(t)$:

$$\frac{c}{N} \sum_{i=1}^n \sum_{j=1}^n \theta_{i,j} S_i I_j = \frac{k(t)\beta}{N} \sum_{i=1}^n S_i \sum_{j=1}^n I_j \quad (12)$$

where

$$k(t) = \frac{c \sum_{i=1}^n \sum_{j=1}^n \theta_{i,j} S_i I_j}{\beta \sum_{i=1}^n S_i \sum_{j=1}^n I_j} \quad (13)$$

This quantity, $k(t)$ only differs slightly from the previous definition of unity beta, $\hat{\beta}(t)$, where it is the scaled ratio of the number of new infections in the homogenous model to the number of new infections in the age-structured model with an equivalent transmission rate. Interpretation is the same as $\hat{\beta}(t)$ above.

2.1 R_0 of the age-structured: using Next Generation Matrix method

To calculate R_0 in the age-structured model, we use the Next Generation Matrix (NGM) method outlined in Diekmann et al 1990. To do so, we follow the steps from the Bjornstad book:

1. Define the infected states, $X = (I_1, I_2, \dots, I_n)$ for ages $i = 1, 2, \dots, n$
2. Define the new infection terms, $\mathcal{F}(X) = \left[\frac{\beta}{N} \sum_{i=j}^n \theta_{i,j} I_j S_i \right]$ for ages $i = 1, 2, \dots, n$
3. Define all losses out of infected states, $\mathcal{V}^-(X) = \left[(\gamma + \mu) I_i \right]$ repeated for all ages.
4. Define all gained transfers between infected states, $\mathcal{V}^+(X) = \left[0 \right]$ repeated for all ages.
5. Define the total losses out of infected states, $\mathcal{V}(X) = \mathcal{V}^-(X) - \mathcal{V}^+(X) = \left[(\gamma + \mu) \right]$.
6. Compute the jacobians of F and V . We do not evaluate at the disease-free equilibrium, because we will also use this evaluated at to calculate R_t . The jacobian of V , $v = \left[\frac{\partial \mathcal{F}_i}{\partial X_j} \right]$, will be $v = (\gamma + \mu)$ along the diagonal and zero elsewhere. The jacobian of F is

$$f = \left[\frac{\partial \mathcal{F}_i}{\partial X_j} \right] = \frac{\beta}{N} \begin{bmatrix} \theta_{1,1} S_1 & \theta_{1,2} S_1 & \cdots & \theta_{1,n} S_1 \\ \theta_{2,1} S_2 & \theta_{2,2} S_2 & \cdots & \theta_{2,n} S_2 \\ \vdots & \vdots & \ddots & \vdots \\ \theta_{n,1} S_n & \theta_{n,2} S_n & \cdots & \theta_{n,n} S_n \end{bmatrix}$$

7. Compute the next generation matrix, $K = f v^{-1}$. Because v is diagonal, then v^{-1} is another diagonal matrices with elements $1/(\gamma + \mu)$. Then, R_0 is the largest eigenvalue of K , evaluated at the disease-free equilibrium (or with $S(t)$ for R_t).